

EXPLORATORY DATA ANALYSIS OF SAAS PRODUCT USAGE & CUSTOMER RETENTION

Subscription Behaviour, Product
Usage & Customer Retention



TOOLS USED

Python, Pandas, Matplotlib, Seaborn & Plotly

PRESENTED BY

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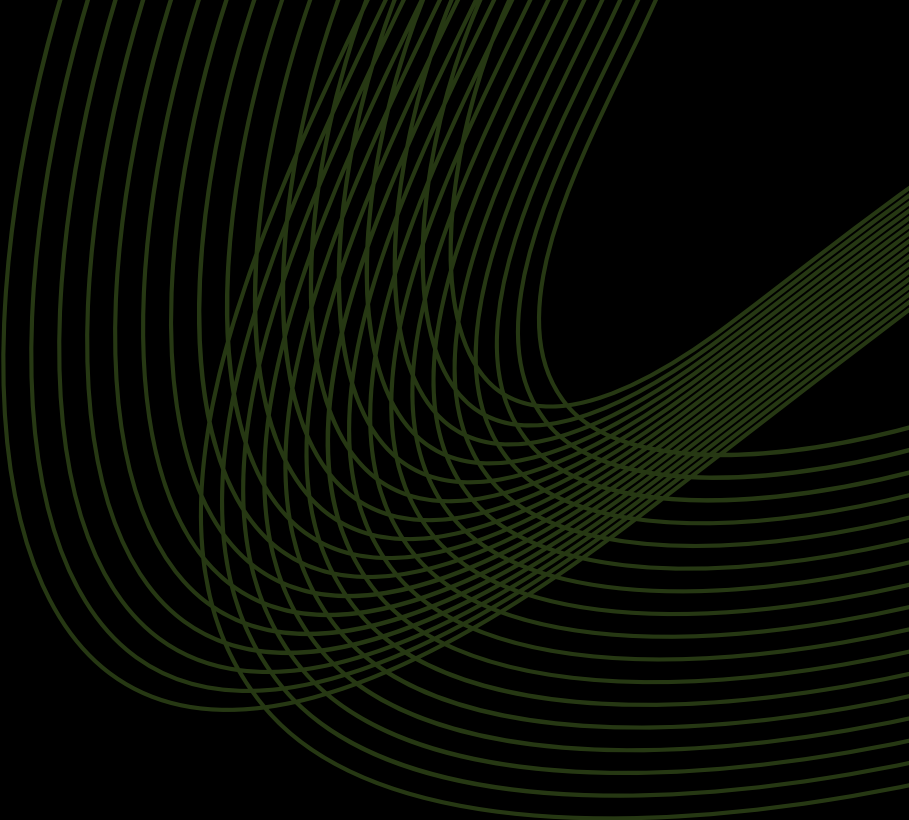


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PROBLEM STATEMENT & OBJECTIVE

Problem Statement:

The Amazon platform hosts millions of products across diverse categories. With massive variations in purchases, ratings & reviews, it becomes challenging to understand consumer trends, customer preferences & what drives product success.

Objective:

To explore the Amazon consumer behaviour dataset and uncover patterns, insights, and trends that help businesses and sellers make data driven decisions.



EDA WORKFLOW

For this analysis, a structured workflow was followed, involving data collection, understanding, cleaning, exploration, and summarization of insights to allow a clear understanding of the dataset and its trends.

01

Data Collection

- Gathered the Google Play Store dataset from Kaggle

02

Data Understanding & Anomaly Detection

- Looked at data distributions
- Found missing values, outliers, and unusual patterns

03

Data Cleaning & Treatment

- Fixed missing or incorrect values
- Standardized formats for consistency

04

Exploratory Analysis

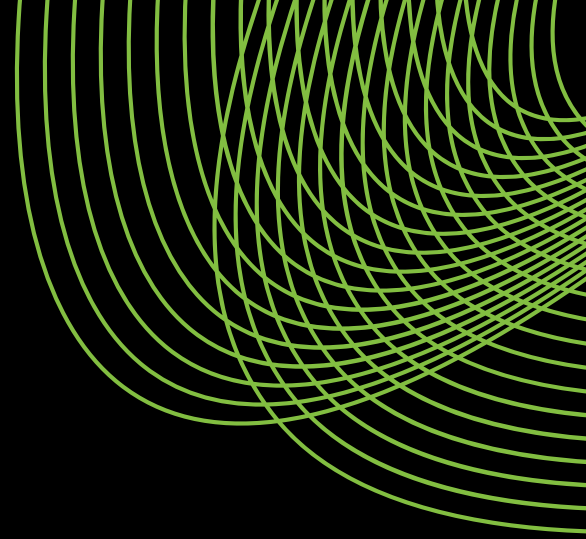
- Univariate Analysis: e.g Purchase frequency, Search Method, Reviews
- Bivariate Analysis: Age vs Purchase Frequency
- Multivariate Analysis: Shopping Satisfaction vs Purchase Frequency vs Age Groups

05

Insights & Reporting

- Summarized patterns and trends
- Highlighted key findings for developers and businesses

KEY QUESTIONS EXPLORED



- How do user demographics (age and gender) influence shopping behavior and engagement patterns on the platform?
- Which stages of the shopping journey (browsing, cart, checkout) contribute most to drop-offs and missed conversions?
- Do occasional shoppers present a greater growth opportunity compared to already frequent buyers?
- How effective are product recommendations in influencing purchase decisions, and where do they fall short?
- What role does pricing perception play in cart abandonment and cross-platform comparison behavior?
- How do customer support responsiveness and delivery reliability impact overall satisfaction and trust?
- How do reviews and post-purchase experiences influence future purchase intent and platform credibility?
- Which satisfaction drivers most strongly contribute to repeat purchases and long-term loyalty?



DATA OVERVIEW

The dataset provides insights into Amazon consumer behavior, covering demographics, purchasing and browsing patterns, product search behavior, cart interactions, and the influence of reviews and personalized recommendations. It also captures overall shopping satisfaction and user feedback for service improvement.

Data Source: Kaggle

Dataset Size

602

Records

23

Features

Purchase Diversity

29

Product Categories

4

Search Methods

5

Satisfaction Indices



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Timestamp	Date	Response time
Age	Numerical (Discrete)	Age of the user
Trending_date	Categorical	Date the video trended on YouTube
Purchase_Frequency	Categorical	How often the user buys on Amazon
Purchase_Categories	Categorical	Types of products the user buys
Personalized_Recommendation_Frequency	Categorical	How often purchases are influenced by recommendations
Browsing_Frequency	Categorical	How often the user browses Amazon
Product_Search_Method	Categorical	How the user searches for products
Search_Result_Exploration	Categorical	Whether the user explores multiple pages of results
Customer_Reviews_Importance	Categorical	Importance of reviews in purchase decisions
Add_to_Cart_Browsing	Categorical	Whether the user adds items to the cart while browsing
Cart_Completion_Frequency	Categorical	How often cart items are purchased
Cart_Abandonment_Factors	Categorical	Reasons for leaving items in the cart
Saveforlater_Frequency	Categorical	Frequency of using "Save for Later"
Review_Left	Categorical	Whether the user has left a product review
Review_Reliability	Categorical	Trust in product reviews
Review_Helpfulness	Categorical	Usefulness of other customers' reviews

DATA OVERVIEW

Continued..

Dataset Features	Type	Feature Description
Personalized_Recommendation_Frequency	Categorical	How often user receives personalized recommendations
Recommendation_Helpfulness	Categorical	USefulness of personalized recommendations
Rating_Accuracy	Categorical	Relevance of recommendations
Shopping_Satisfaction	Categorical	Overall satisfaction with Amazon
Service_Appreciation	Categorical	Most Appreciated Amazon Services
Improvement_Areas	Categorical	Suggested areas for Amazon to improve

DATA QUALITY CHALLENGES & ANOMALIES

Few inconsistencies were found in the dataset, which could have affected the analysis if left unaddressed.

DATA ANOMALIES

- **Columns requiring cleaning or type conversion:** timestamp, Gender, Purchase_Categories, Product_Search_Method, Search_Result_Exploration, Customer_Reviews_Importance, Add_to_Cart_Browsing, Cart_Abandonment_Factors, Review_Left, Service_Appreciation, Improvement_Areas, Rating_Accuracy, Shopping_Satisfaction
- **Personalized_Recommendation_Frequency** appears twice with different types (str and int64) resolve duplication (rename or remove).
- **Columns with notable anomalies or invalid entries:**
 - Improvement_Areas & Service_Appreciation (has an entry like `.`)
- Two missing values in **Product_Search_Method** column.



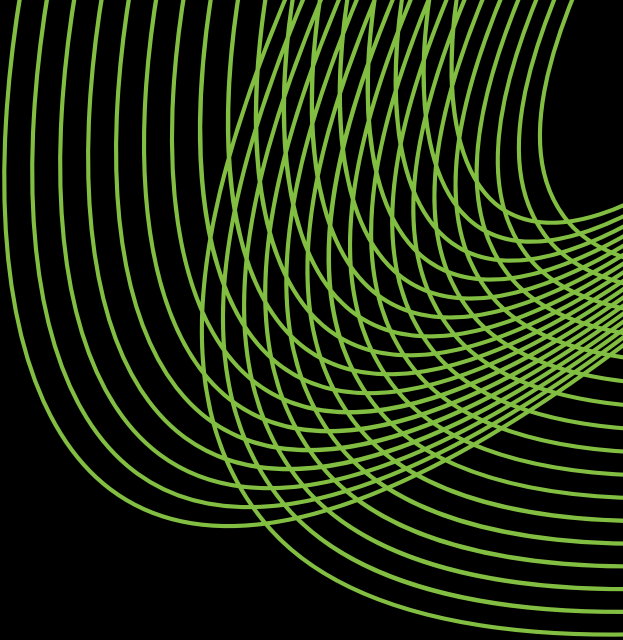
DATA CLEANING & TREATMENT

Inconsistent and missing values were addressed, and key features were cleaned and standardized for analysis.

DATA CLEANING SUMMARY

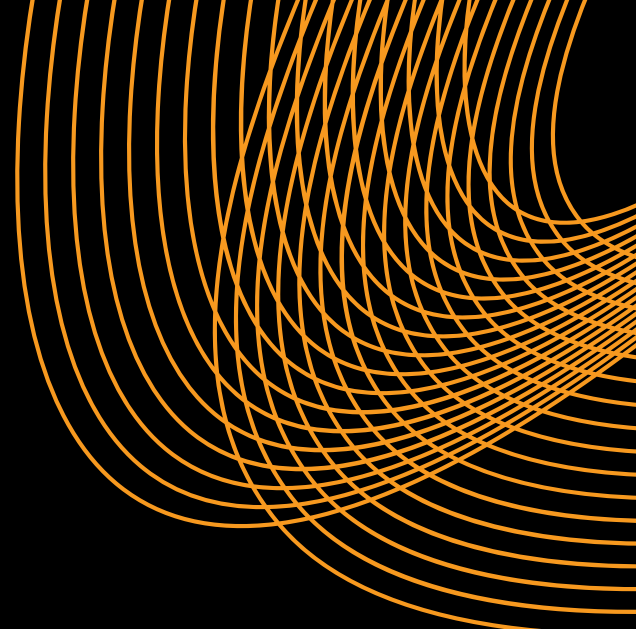
- Features such as **timestamp**, **Gender**, **Purchase_Categories**, **Product_Search_Method**, **Search_Result_Exploration**, **Customer_Reviews_Importance**, **Add_to_Cart_Browsing**, **Cart_Abandonment_Factors**, **Review_Left**, **Service_Appreciation**, **Improvement_Areas**, **Rating_Accuracy**, **Shopping_Satisfaction** were standardized with respect to data type conversion.
- **Improvement_Areas** & **Service_Appreciation** were cleaned to handle invalid entries (. replaced with mode).
- **Personalized_Recommendation_Frequency** were resolved and renamed as:
 - '**Personalized_Recommendation_Frequency**' to **Number_of_times_Personalized_Recommendation_Received**
 - '**Personalized_Recommendation_Frequency**' to **Purchase_made_on_Personalized_Recommendation**
- Missing values in **Product_Search_Method** were replaced with mode.





INSIGHTS





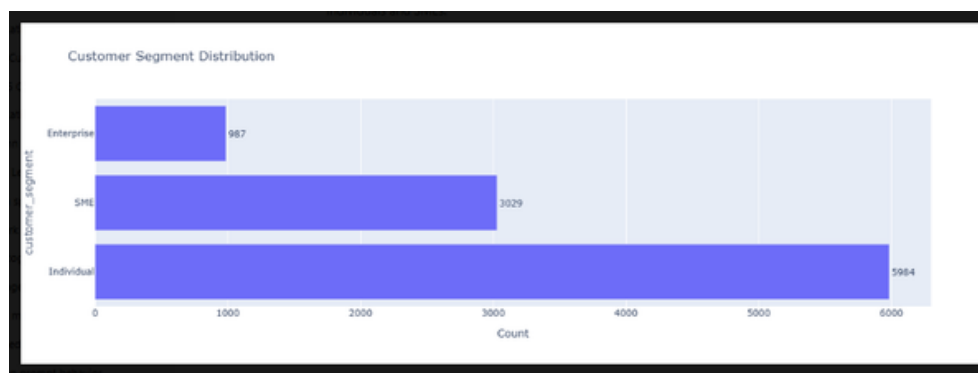
Customer Profile & Account Overview



Customer base is heavily skewed toward Individuals, while Enterprise adoption remains limited

58.5%

Of the Customers are Females



Key observations

- Individual customers make up the largest share of the user base, far exceeding both SME and Enterprise segments.
- Enterprise customers represent a very small portion of total users, indicating low penetration compared to other segments.

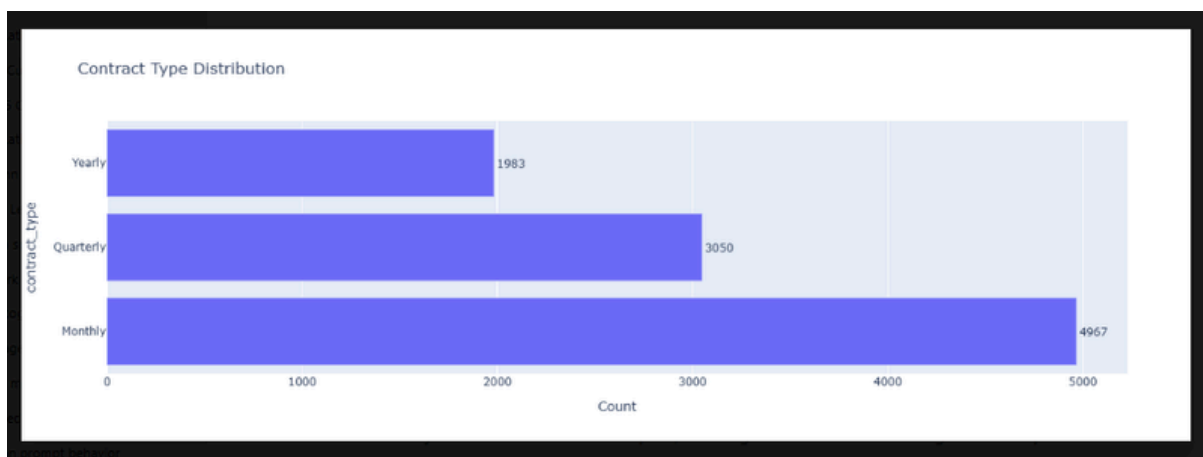
Business Insights

- Strong Individual adoption suggests the product is well-positioned for self-serve usage, but long-term revenue growth may be constrained without deeper SME and Enterprise expansion.
- The limited Enterprise presence highlights a need to reassess enterprise-specific offerings—such as pricing, features, or sales strategy—to unlock higher-value customer growth and improve revenue stability.

Short-term contracts dominate customer subscriptions, limiting long-term revenue stability

67%

Of the Customers are Young Adults



Key observations

- Monthly contracts are the most common subscription type, significantly outnumbering both Quarterly and Yearly plans.
- Yearly contracts have the lowest adoption, indicating limited commitment to long-term subscriptions.

Business Insights

- Heavy reliance on monthly contracts increases churn exposure and makes revenue less predictable over time.
- Encouraging upgrades to Quarterly and Yearly plans through pricing incentives or value bundling could improve retention and stabilize long-term revenue.

Customer tenure is broadly distributed, indicating steady retention beyond early lifecycle stages

33.72%

Customers Make Purchases on a Monthly Basis



Key observations

- Customers are spread relatively evenly across tenure months, with no sharp drop-off in the early subscription period.
- A substantial portion of users remain active well beyond the first year, extending toward long-term tenures.

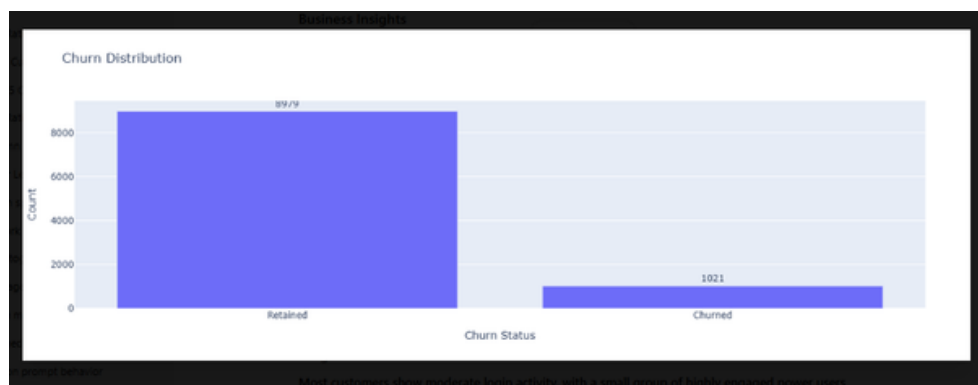
Business Insights

- The absence of early tenure drop-offs suggests effective onboarding and early product adoption.
- Since customers persist across long tenures, improving engagement and expansion strategies for mid- to long-term users could further strengthen lifetime value and reduce long-term churn risk.

Overall churn remains low, indicating strong baseline customer retention

41.36%

Customers Frequently Return to Amazon Each Week

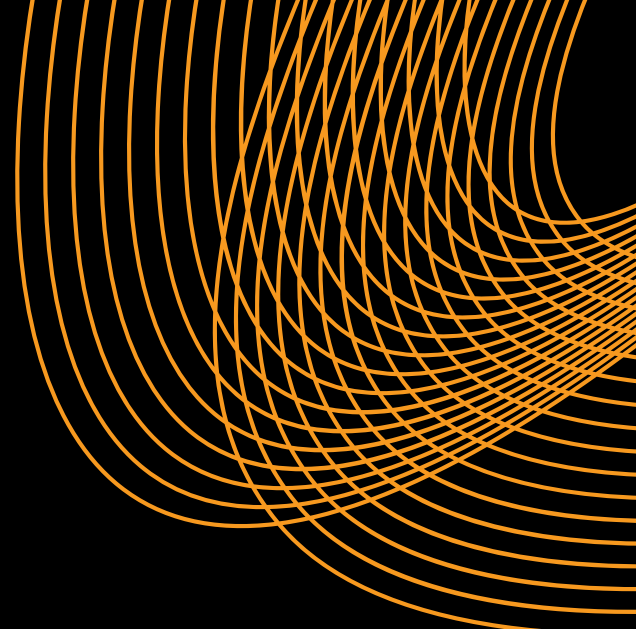


Key observations

- The vast majority of customers are retained, with churned users representing only a small fraction of the total customer base.
- There is a clear imbalance between retained and churned customers, showing that most users continue their subscriptions.

Business Insights

- Low overall churn suggests the product delivers sufficient ongoing value for most customers.
- Despite strong retention at an aggregate level, targeted analysis is needed to identify high-risk subgroups, as small churn segments can still have an outsized impact on long-term revenue and growth.



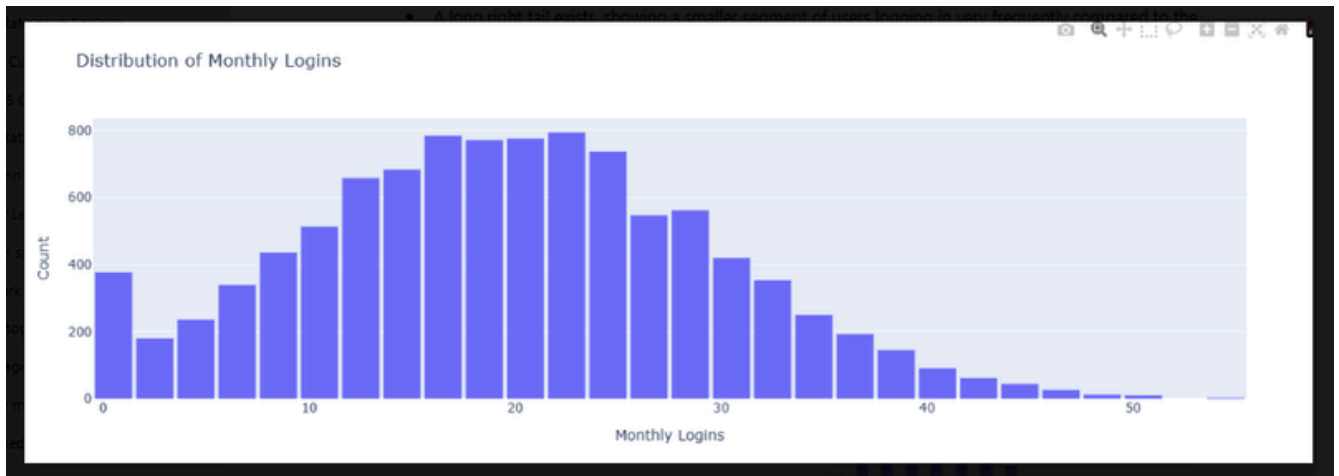
Product Adoption & Usage Health



Most customers show moderate login activity, with a small group of highly engaged power users

50.5%

Cart Drop-Offs Occur Frequently



Key observations

- Monthly logins are concentrated in the mid-range, indicating consistent but not daily usage for most customers.
- A long right tail exists, showing a smaller segment of users logging in very frequently compared to the majority.

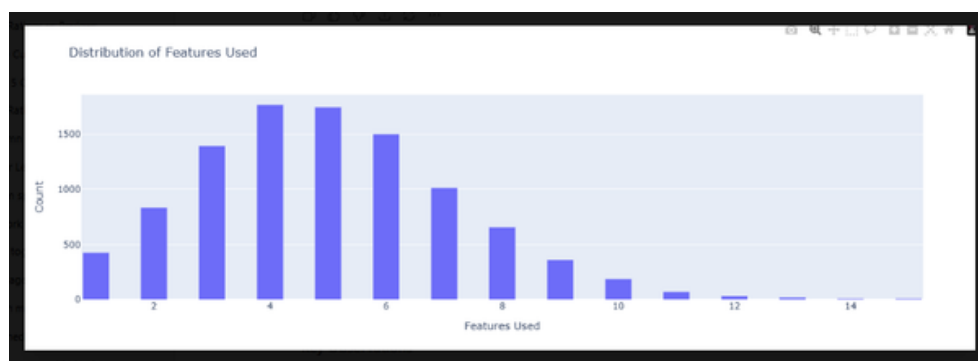
Business Insights

- The concentration of users in the moderate activity range suggests healthy baseline engagement but room to deepen habitual usage.
- Highly active users likely represent power users or high-value accounts, making them strong candidates for upsell, advocacy, and referral-driven growth strategies.

Most customers engage with a limited set of core features, while deep feature adoption remains rare

42.35%

Portion of Consumers Prioritize Price Over Brand Loyalty



Key observations

- Feature usage is heavily concentrated around a small number of features, with most customers using roughly 3–6 features.
- Only a small minority of users adopt a wide range of features, creating a long tail of advanced usage.

Business Insights

- Concentrated usage around core features suggests these drive primary product value, but many customers may not be fully realizing the platform's capabilities.
- Improving feature discovery, onboarding, and in-product guidance could expand feature adoption and increase customer stickiness, helping reduce long-term churn risk.

Low engagement intensity strongly aligns with churn, while retained users show broader and more frequent product usage



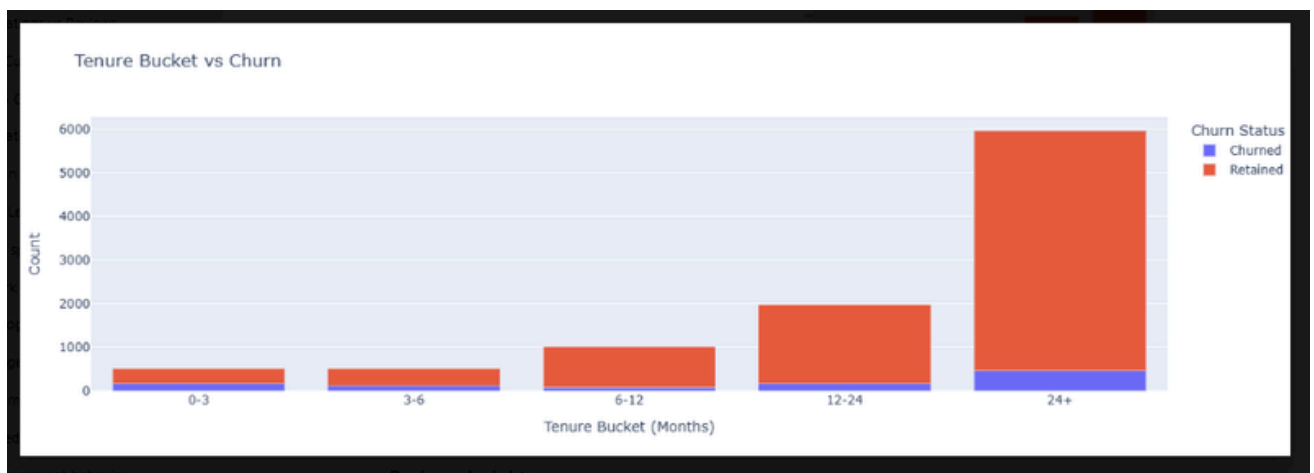
Key observations

- Churned customers are heavily concentrated at lower monthly logins and fewer features used.
- Retained customers are more prevalent at higher login frequencies and higher feature adoption levels.

Business Insights

- Customers who fail to build regular usage habits and explore multiple features are significantly more likely to churn.
- Driving early engagement through activation nudges, feature discovery, and usage-based onboarding can materially reduce churn risk by moving users into higher-engagement zones.

Churn risk is front-loaded in early tenure, while long-tenured customers show strong retention stability

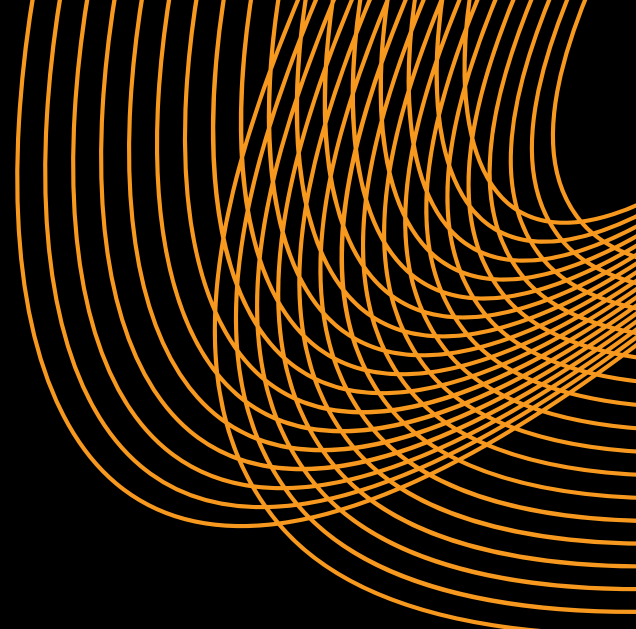


Key observations

- Churn is proportionally higher in the early tenure buckets (0-3 and 3-6 months) compared to later stages.
- Customers with longer tenure, especially 24+ months, are overwhelmingly retained with minimal churn presence.

Business Insights

- The first few months represent the most critical window for preventing churn, making onboarding and early activation essential.
- Once customers cross long-term tenure thresholds, retention stabilizes, indicating that efforts to push users past early lifecycle stages can significantly improve lifetime value.



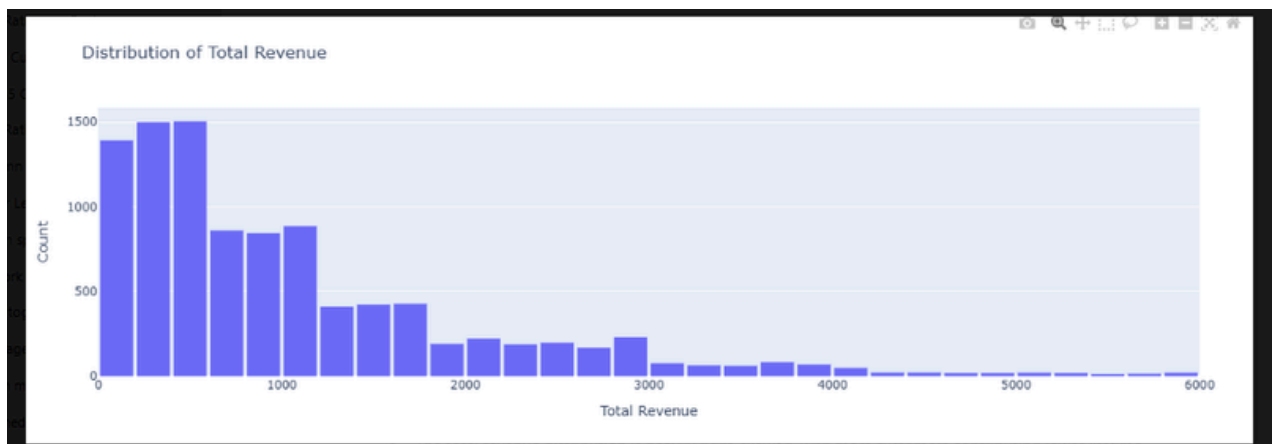
Revenue & Pricing Sensitivity



Revenue is concentrated among a small group of high-value customers, while most generate modest lifetime value

16.94%

customers report that online reviews affect their buying choices



Key observations

- The total revenue distribution is heavily right-skewed, with most customers contributing lower lifetime revenue.
- A long tail of customers generates significantly higher revenue, indicating the presence of a small high-value segment.

Business Insights

- A minority of customers likely drive a disproportionate share of total revenue, making their retention critical to overall business performance.
- Identifying the behaviors and characteristics of these high-value customers can inform targeted retention, upsell, and expansion strategies to maximize lifetime value.

Early payment failures show a visible association with increased churn

51.5%

Of the Customers are Review Contributors



Key observations

- Most retained customers have zero or only one payment failure.
- Churned customers are disproportionately represented as the number of payment failures increases.

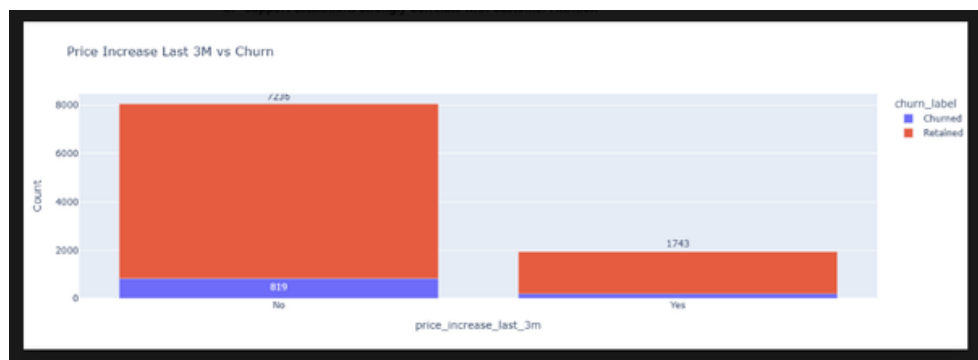
Business Insights

- Payment friction appears to be a key driver of involuntary churn.
- Proactive interventions such as payment retry logic, early alerts, and alternative payment options could significantly reduce avoidable customer losses.

Price Increases Show Minimal Impact on Customer Churn

33.72%

Customers Make Repeat Purchases
Several Times a Month



Key observations

- Churn levels among customers who experienced a price increase remain relatively low compared to retained customers.
- The majority of churn comes from customers who did not face a recent price increase.

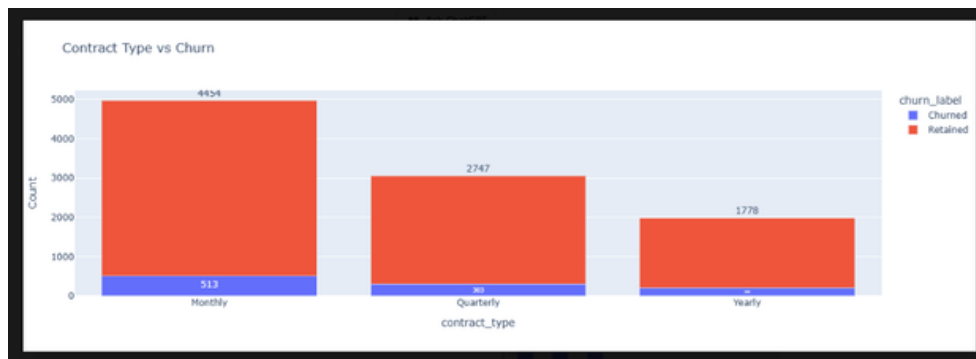
Business Insights

- Recent price adjustments do not appear to be a major churn trigger, suggesting strong perceived product value.
- This indicates potential room for strategic pricing optimization without significantly harming retention.

Short-Term Contracts Drive Higher Churn Risk Compared to Long-Term Plans

33.72%

Customers Make Repeat Purchases
Several Times a Month

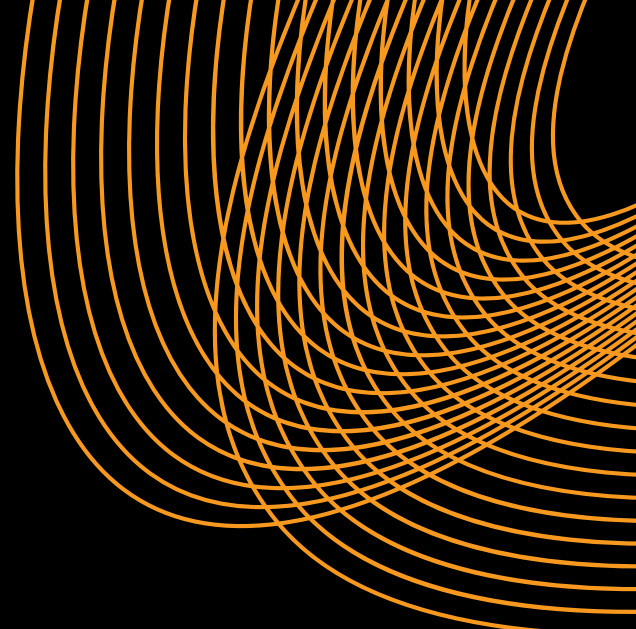


Key observations

- Monthly contracts show the highest churn volume compared to Quarterly and Yearly plans.
- Yearly contracts exhibit the lowest churn levels, with significantly stronger retention.

Business Insights

- Customers on short-term (Monthly) plans are more prone to churn, indicating lower commitment levels.
- Incentivizing upgrades to longer-term contracts (Quarterly/Yearly) could materially improve retention and revenue stability.



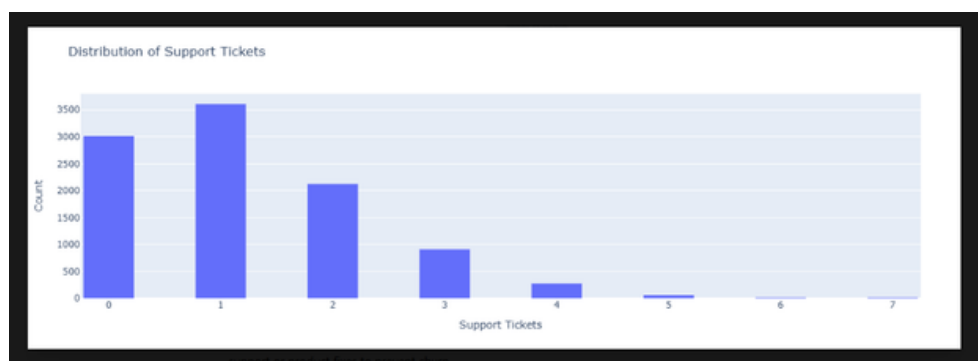
Customer Support & Experience Quality



Customer Support Demand Is Concentrated Among Low-Ticket Users

20.3%

Customers Are Responsive to Personalized Buying Suggestions



Key observations

- The majority of customers raise 0-1 support tickets, with counts declining sharply beyond 2 tickets.
- Very few customers generate high ticket volumes (4+), indicating a small segment of heavy support users.

Business Insights

- Most customers experience minimal friction, suggesting overall product stability.
- A small high-ticket segment may represent at-risk or high-maintenance customers, requiring proactive support or product fixes to prevent churn.

Longer Resolution Times Combined with Frequent Support Tickets Significantly Increase Churn Risk—Especially Among Monthly Contracts



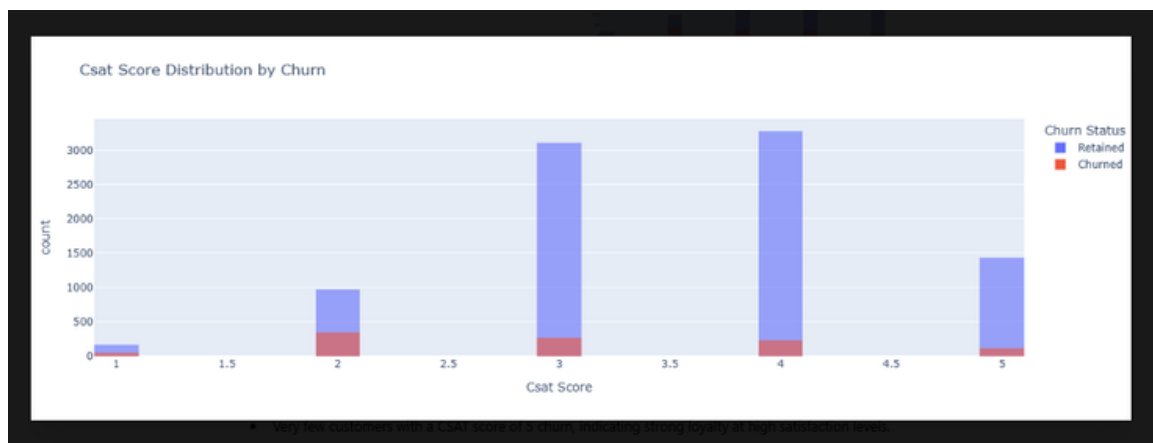
Key observations

- Churn correlates with higher support ticket counts and longer resolution times, with Monthly customers showing the highest sensitivity, Yearly customers the lowest, and Quarterly in between.
- Yearly customers retain better than Monthly or Quarterly even at similar support volumes, highlighting contract length's buffering effect on churn.

Business Insights

- Support friction—both ticket volume and slow resolution—is a clear churn driver, especially for Monthly plans with low commitment.
- Faster resolution and proactive support (SLAs, escalation, early intervention) can reduce churn significantly, potentially without changing pricing or features.

Strong retention signal here. Very clean.



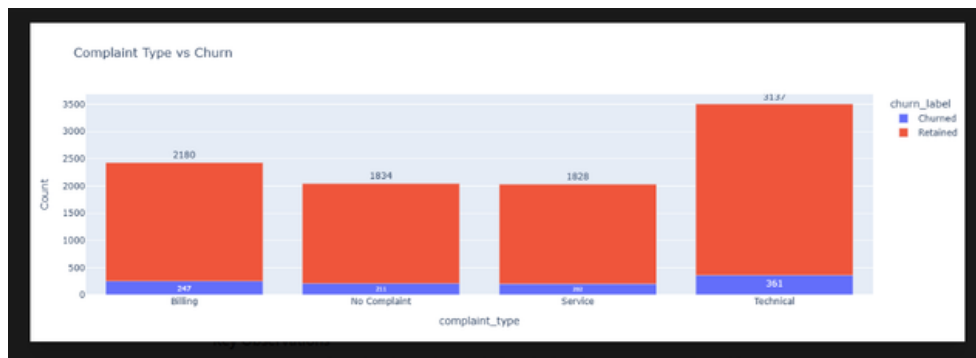
Key observations

- Customers with low CSAT (1-2) have much higher churn, while the majority of retained customers score 3-5, with very few churning at 5.
- Churn steadily declines as satisfaction increases, showing strong loyalty at high CSAT scores.

Business Insights

- Customer satisfaction is a leading churn indicator; even moderate dissatisfaction (2-3) increases retention risk.
- Raising CSAT by one point and using low scores (≤ 2) as early-warning triggers can meaningfully reduce churn through proactive outreach.

Technical Complaints Drive the Highest Churn Risk Compared to Other Complaint Types

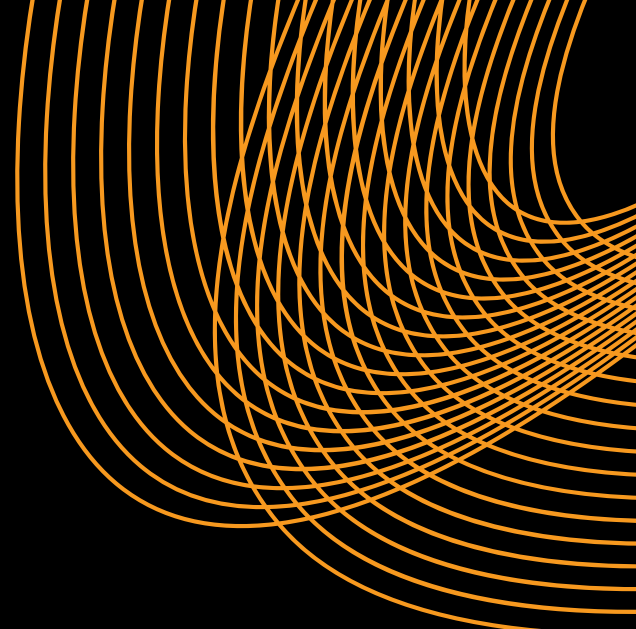


Key observations

- Technical issues drive the highest churn, followed by billing and service complaints; even customers with no complaints churn, though less frequently.
- Technical problems are the most critical friction point affecting retention.

Business Insights

- Prioritizing technical issue resolution through product stability, bug fixes, and proactive monitoring can significantly reduce churn.
- While billing/service complaints matter, differentiated support strategies and holistic retention efforts are needed for "no complaint" churners.



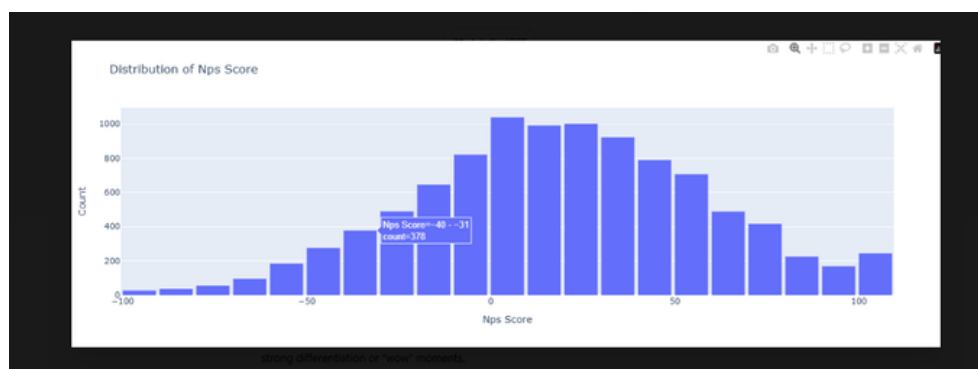
Engagement, Loyalty & Advocacy



Customer Sentiment Skews Neutral-to-Positive, but a Meaningful Detractor Base Signals Churn Risk

34.7%

Average Satisfaction Indicates Room for Improvement



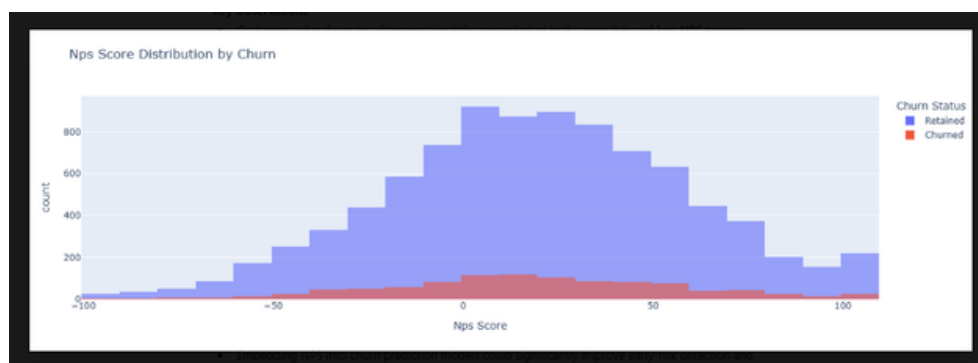
Key observations

- The NPS distribution is concentrated in the mid-range (0-40), indicating a large passive customer base, with fewer extreme high promoter scores (80-100), forming a broad bell-shaped curve rather than a strong right skew.
- There is a clear cluster of negative scores (below 0), showing a meaningful detractor segment alongside the dominant passive group.

Business Insights

- The large passive base is a major conversion opportunity, while the detractor segment signals potential churn risk — especially if linked to low usage, support issues, or pricing changes.
- The limited dominance of strong promoters suggests the product is satisfactory but not highly differentiated, meaning improvements in onboarding, feature adoption, and support experience could shift the curve right and increase retention and referrals.

Low NPS Scores Strongly Correlate with Churn, While Promoters Show Higher Retention Stability



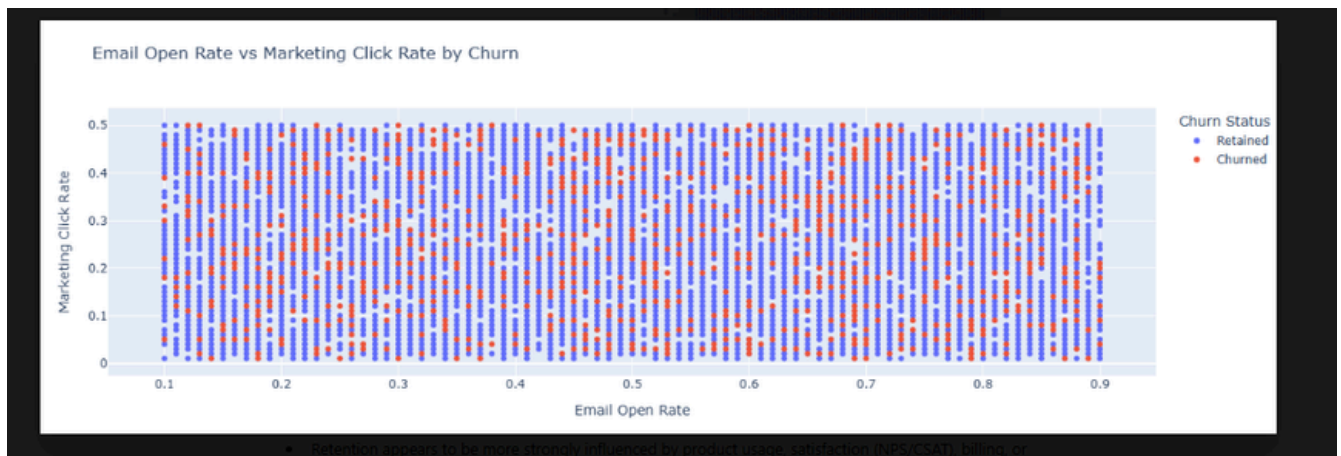
Key observations

- Customers who churn are mostly in the negative and low NPS ranges, while retained customers dominate mid to high NPS scores (20–60+), with very high scores (80–100) almost entirely retained.
- The mid-range (0–30) shows overlap between churned and retained customers, highlighting a significant passive segment present in both groups.

Business Insights

- NPS is a strong leading indicator of churn; detractors (<0) need proactive intervention, and passives (0–30) are a key conversion opportunity to reduce churn.
- Promoters are high-retention assets and referral drivers, and embedding NPS into churn models can enhance early-risk detection and retention strategies.

Marketing Engagement Alone Does Not Clearly Differentiate Churned vs Retained Customers



Key observations

- Retained and churned customers overlap heavily in email open rates ($\approx 10\%$ – 90%) and click rates ($\approx 0\%$ – 50%), with no clear low-engagement clustering among churned users.
- High engagement does not guarantee retention, as churn occurs even among users with strong open and click rates.

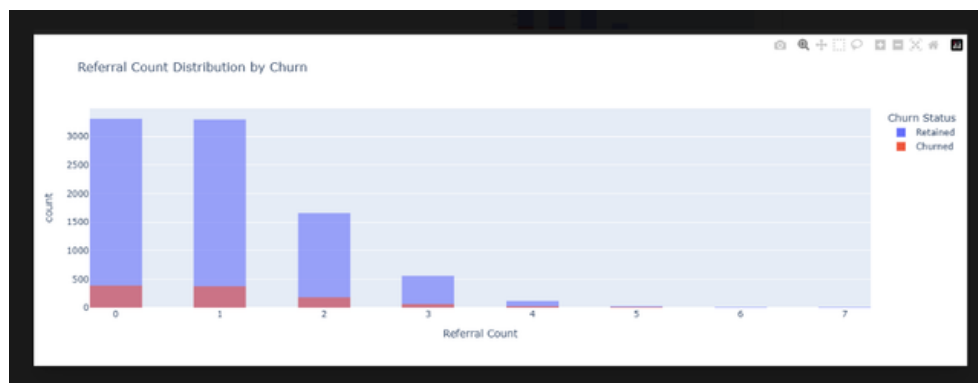
Business Insights

- Email engagement alone is a weak predictor of churn; retention is more influenced by product usage, satisfaction (NPS/CSAT), billing, or support experience.
- Marketing metrics should support broader churn models, and churn among highly engaged users may signal messaging misalignment, feature gaps, or pricing sensitivity.

Higher Referral Activity Strongly Correlates with Customer Retention and Lower Churn Risk

36.87%

Customers Desire Better Support Experience



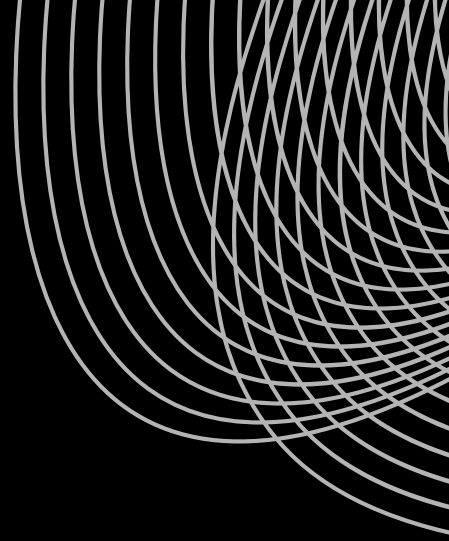
Key observations

- Customers with 0-1 referrals account for the vast majority of churn, while churn drops sharply as referral count increases.
- Customers with 3+ referrals are overwhelmingly retained, indicating a strong inverse relationship between referral activity and churn risk.

Business Insights

- Referral behavior is a powerful proxy for engagement and loyalty — customers who actively refer are significantly less likely to churn, suggesting advocacy reflects deep product value realization.
- Driving referral participation early in the lifecycle can serve as both a growth and retention lever, turning passive users into long-term, high-LTV advocates while helping identify low-referral customers as churn-risk segments.

BUSINESS / DEVELOPER TAKEAWAYS



- **Female-Led Usage with Inclusive Design Needs:** Shopping behaviour is more aligned with female users, but a sizable undisclosed segment makes gender-neutral personalization essential for scale.
- **Young Adults Drive Core Demand:** Young adults dominate the user base, making them the primary target for products, messaging & promotions, while simplified UX and trust signals can help retain older users.
- **Occasional Shoppers Represent the Biggest Growth Opportunity:** Targeted reminders, deals, and subscription incentives can convert occasional shoppers more effectively than focusing only on already frequent buyers.
- **High Browsing, Low Conversion Signals Untapped Potential:** Strong browsing activity highlights opportunities to convert interest into purchases through improved discovery, timely nudges, and smarter recommendations.
- **Checkout Friction Is a Key Drop-Off Point:** Reducing uncertainty at checkout through price clarity, trust signals, and cart reminders can significantly improve completion rates.



BUSINESS / DEVELOPER TAKEAWAYS

- **Price Sensitivity Drives Abandonment:** Competitive pricing, price-match cues, and limited-time offers are effective in converting undecided users who compare across platforms.
- **Loyalty Should Focus on Purchase Frequency, Not Age:** High-frequency shoppers respond well to loyalty benefits and faster checkout, while low-frequency shoppers need friction reduction and reassurance to complete purchases.
- **Reviews Influence Decisions, when curated well:** Clear, concise, and highlighted “most helpful” reviews influence moderately review-sensitive users without overwhelming them.
- **Post-Purchase Experience Fuels Organic Reviews:** Improving satisfaction after purchase encourages higher-quality reviews and greater participation without relying heavily on incentives.
- **Recommendation Quality Determines Real Impact:** Personalization only works when relevance is high. Improving accuracy and timing can turn occasional influence into consistent purchasing behavior.
- **Satisfaction Is the Strongest Loyalty Multiplier** Small improvements in UX, pricing, customer support and delivery reliability can yield outsized gains in repeat purchases and long-term trust.



CHALLENGES & OPPORTUNITIES

Limitations

- The dataset is survey-based and may not fully reflect actual or current Amazon consumer behavior trends.
- Respondents represent a limited subset of Amazon users, potentially introducing bias toward more active or engaged shoppers.
- Timestamped responses provide only a snapshot in time, limiting long-term trend analysis.

Future Work / Opportunities:

- Perform sentiment analysis on customer reviews to understand consumer perception and satisfaction.
- Conduct time-series analysis on purchase and browsing patterns to study how consumer engagement evolves over time.
- Compare frequent buyers with infrequent buyers to identify factors driving purchase behavior.
- Analyze the impact of product categories, personalized recommendations and review metrics on shopping decisions.



CONCLUSION

This analysis of Amazon consumer behavior provided a comprehensive understanding of purchasing patterns, browsing habits, and engagement with platform features. Key insights and trends were identified, demonstrating how data-driven analysis can inform strategic decisions for sellers, marketers and platform managers. Future analysis could include sentiment analysis of customer reviews, time-series tracking of purchase and browsing patterns and comparisons between frequent and infrequent buyers to uncover deeper insights.

This analysis was conducted using Python, Pandas, Matplotlib, Seaborn and Plotly.

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